Sogakope Rentals

OFFICIAL ENGINEERING & PLATFORM ARCHITECTURE SPECIFICATION

Document Version: 1.2
Status: Production Ready
Region: South Tongu District

CORE FRAMEWORK	DATABASE & AUTH	DEPLOYMENT	PRIMARY STYLING
Next.js (App Router)	**Supabase (PostgreSQL)**	**Vercel Edge Platform**	**Tailwind CSS + Lucide**

1. Platform Scope & Objective

Sogakope Rentals is a hyper-local accommodation directory serving tenants, health professionals, educators, and tertiary students across the South Tongu District of Ghana. The application removes unverified intermediaries by centralizing rental listings with transparent pricing in Ghana Cedis (GH₵), verified amenity audits (borehole water, postpaid/prepaid meters), and direct communication paths to property owners.

2. Data Schema: Public Listings (`properties`)

Field Name	SQL Data Type	Constraints	Functional Role
`id`	uuid	PRIMARY KEY	Unique property record identifier
`title`	text	NOT NULL	Listing headline (e.g. "Apartment Near district hospital")
`price`	numeric	NOT NULL	Monthly rent rate in Ghana Cedis (GH₵)
`location`	text	NOT NULL	Landmark context (e.g., Texas, Comboni, Cisneros area)
`category`	text	NULLABLE	Self Contain, Chamber & Hall, Student Hostel, Single Room
`landlord_phone`	text	NOT NULL	Direct telephone and WhatsApp contact digits
`status`	text	DEFAULT 'pending'	Visibility state: 'pending', 'approved', or 'rejected'

Feed Filtering Policy: Feed index queries execute with `eq('status', 'approved')` to ensure unapproved and rejected submissions do not leak into consumer search results.

3. Incident & Report Management Schema (`property_reports`)

Field Name	SQL Data Type	Constraints	Description
`id`	uuid	PRIMARY KEY	Auto-generated UUID primary identifier
`property_id`	uuid / text	FOREIGN REF	Associated listing subject to visitor review
`reason`	text	NOT NULL	Primary category ("Incorrect price", "Unavailable", "Scam")
`status`	text	DEFAULT 'pending'	Lifecycle flag: 'pending', 'reviewed', or 'resolved'
`contact_phone`	text	NULLABLE	Reporter telephone follow-up contact details

4. Security Architecture & Administrative Route Shielding

Default administrative URL paths are actively targeted by automated brute-force scripts. Sogakope Rentals deploys a strict two-layer defense against administrative scanning:

```
// Intercepting /admin to eliminate reconnaissance visibility export async function GET()
```

> **Zero Information Leakage:** The `/admin` handler intentionally omits standard Next.js error layout wrappers, returning an unstyled raw HTTP 404 response. Legitimate administrative controls operate exclusively through an obfuscated private path backed by authenticated session validation.

5. Deployment, Health Verification & Release Cadence

1. **Continuous Integration Audits**
   Prior to release, execute lint and type-checking checks via `npm run typecheck` and `npm run build` to verify strict TypeScript compliance across all route files.
2. **Immutable Version Tracking**
   Stage modifications, declare semantic commit flags (`feat:` , `fix:` , `docs:`), and push changes directly to the remote `origin/main` repository branch.
3. **Edge Production Distribution**
   Automated Vercel pipeline checks run integration builds, invalidating the edge cache and serving updated assets globally under zero-downtime rolling deploys.